

B-INFOHUB BRIDGE v1.1 -- Briggs & Stratton GC-1032 RS-485 bridge with InfoHub passthrough

POWER IN: 12 V from the generator harness. Q1 reverse-polarity (carries the InfoHub passthrough too, 4 A part). D2 TVS clamps starter load-dump. F2 fuses the InfoHub separately.

3.3 V RAIL: AP63203WU-7 sync buck, 3.8-32 V in, fixed 3.3 V 2 A (circuit from b-hydro carrier v2.1). Rev 1.1: one rail. The 5 V rail only fed the DevKit's linear 3.3 V regulator, which is gone.

power flags

PORT A -- GENERATOR CONTROLLER. We are Modbus MASTER here. Slave 10, 9600 8E1 (measured). RE and DE tied: one GPIO (IO4) drives direction. JP1 fits the 120R terminator.

PORT B -- INFOHUB PASSTHROUGH. We are Modbus SERVER here, impersonating the controller at slave 10 so the InfoHub keeps working. R10/R11 bias the idle pair (rev 1.1).

ESP32-WROOM-32E soldered (rev 1.1; was a socketed DevKitC). Same pads take the -32UE for a U.FL remote antenna. No strapping pin carries a function; IO0 is BOOT only.

EN/BOOT: 10k pull-ups, 1 uF on EN. SW1 RESET, SW2 BOOT (diagonal pads only -- see circuit.py). J6 PROG: 3V3 GND TX RX EN IO0 -- first flash with any 3.3 V USB-serial adapter, OTA after that.

STATUS LEDs: PWR and TX need no GPIO -- TX is driven from DE, so it proves the bridge is actually transmitting, not merely powered. All on 3.3 V, 100R blue/green.

J4 EXT LED: the same anode nodes, for panel LEDs in parallel. FAN (rev 1.1): 12 V via F3, Q2 low-side PWM from IO32, D10 flyback, R13 holds the fan OFF while the ESP is in reset.